IT05IoT091024

(Time: 2 hours)

[Total Marks: 60]

N. B.: (1) **All** questions are **compulsory**.
(2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.
(3) Answers to the **same question** must be **written together**.
(4) Numbers to the **right** indicate **marks**.
(5) Draw **neat labeled diagrams** wherever **necessary**.
(6) Use of **Non-programmable** calculator is **allowed**.

1. Attempt *any two* of the following: **12**

a. Explain the equation of IOT.
b. "Be conservative in what you do, be liberal in what you accept from others". Discuss.
c. What is 6LoWPAN? Enumerate the features of 6LoWPAN?
d. Discuss the need for classless interdomain routing? What is classless interdomain routing?

2. Attempt *any two* of the following: **12**

a. What is sketching? Explain its role in prototyping.
b. Discuss the advantages and disadvantages of open source and close source hardware.
c. Enumerate the different considerations while choosing the right platform for IoT device.
d. Compare Arduino duo and Raspberry Pi.

3. Attempt *any two* of the following: **12**

a. Explain the following non-digital method of
i) Modelling clay ii) E-Poxy putty iii) Sugru iv) non construction set v) Cardboard vi) Extruded Polystyrene
b. Explain the different types of 3D printing.
c. Discuss the legalities associated with scrapping.
d. What is MQTT protocol? Explain in detail.

4. Attempt *any two* of the following: **12**

a. What are the memory limitations while writing embedded code? Explain.
b. Explain different types of memory.
c. With the help of the diagram explain business model canvas.
d. What are the constraints while receiving funding for IoT start-ups?

5. Attempt *any two* of the following: **12**

a. How are printed circuit board are designed? Explain.
b. Discuss the cost consideration as we move to high value manufacturing.
c. Explain the five critical requirements for a sensor common project.
d. What is environmental cost of Internet service for IOT device? What are the solutions?
